

Department of Environmental Science & Engineering

Environmental Noise Assessment & Impact Analysis



Academic Year: 2026 | EIA Survey Camp Group 4

AB-1 Construction Zone | Delhi Technological University

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Introduction & Context

What is this study?

An acoustic monitoring Study to assess noise pollution and its impact generated by the AB-1 Academic Block construction at DTU a designated Silence Zone under Indian law.

Key Metric: L(eq)

Equivalent Continuous Sound Level (dBA) — a logarithmic average ensuring brief high-intensity events are fully captured. Sound energy doubles every +3 dB; perceived loudness doubles every +10 dB.

Instrument: Cesva SC260 Sound Level Meter
(30–137 dBA)

CPCB Ambient Noise Standards

Zone	Day (dBA)	Night (dBA)
Industrial	75	70
Commercial	65	55
Residential	55	45
Silence Zone ★	50	40

★ DTU Campus falls under Silence Zone — the strictest classification.
Any violation >10 dBA enables legal action under Noise Pollution Rules, 2000.



1. INTRODUCTION



Environmental noise pollution is increasingly recognized as a critical environmental issue in urban and semi-urban regions, particularly in areas undergoing rapid infrastructural development.



Noise is defined as "any unwanted or harmful sound that disrupts normal human activities or degrades environmental quality".



Sound itself is a "form of energy produced by vibrating bodies and transmitted through a medium such as air." However, when this sound becomes excessive, irregular, or occurs at inappropriate times, it is classified as noise.

The human ear can perceive sound within a frequency range of approximately **20 Hz to 20,000 Hz**. The sensitivity of the human ear varies across this range, with maximum sensitivity typically observed between **250 Hz and 5000 Hz**. This variation plays an important role in how environmental noise is measured and evaluated.



1.1 MEASUREMENT OF NOISE AND KEY CONCEPTS



Noise levels are quantified using the unit **decibel (dB)**, which is a logarithmic scale used to express sound intensity. This scale allows a wide range of sound levels to be represented in a compact form. However, the human ear does not respond equally to all frequencies; it is more sensitive to mid-frequency sounds and less sensitive to very low and very high frequencies.

To account for this variation in human hearing, different frequency weightings are used in noise measurements:

**dB
(A)**

A-WEIGHTING (dB(A))

This is the most commonly used scale in environmental noise studies. It closely matches the sensitivity of the human ear by emphasizing mid-frequency sounds and reducing the contribution of very low and high frequencies. Hence, it is widely used for assessing traffic noise, industrial noise, and community noise.

**dB
(B)**

B-WEIGHTING (dB(B))

This scale provides moderate weighting between A and C scales. It was designed for medium sound levels but is rarely used in modern noise measurements.

**dB
(C)**

C-WEIGHTING (dB(C))

This scale gives nearly equal importance to all frequencies (almost flat response). It is used for measuring high-intensity sounds such as explosions, machinery noise, and peak noise levels.



Thus, while dB represents the physical measurement of sound, weighted scales like dB(A), dB(B), and dB(C) adjust this measurement to better reflect human hearing, with dB(A) being the most widely used standard in environmental noise assessment.

Different Noise Descriptors

Parameter	Description
Leq	The constant sound level that would produce the same energy as the fluctuating noise over a given time
Lmax	Maximum recorded noise level during the observation period
Lmin	Minimum recorded noise level during the observation period
Ln	Noise level exceeded for n% of the total time, ex. L10, L50, L90
L10	Noise level exceeded for 10% of the time (indicates peak noise)
L50	Median noise level (50% of time) represents the average/typical background noise level, The noise level that is exceeded for 50% of the total measurement time
L90	Background noise level (90% of time) L90 is the level exceeded for 90% of the time. For 90% of the time, the noise level is above this level.
Ldn	24-hour average noise level where night noise (10 PM–6 AM) is penalized by +10 dB

$$Leq = 10 \cdot \log_{10} \left(\frac{1}{n} \sum_{i=1}^n 10^{\frac{L_i}{10}} \right)$$

Introduction & Context

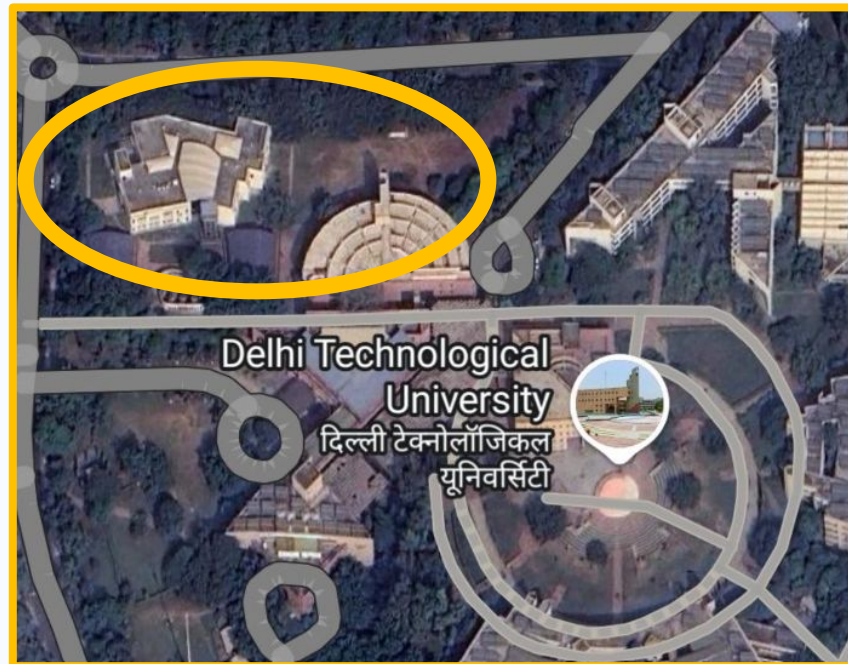
Noise Measurement Equipment Details

Instrument: Cesva SC260 Sound Level Meter
(30–137 dBA)

Instrument	Model	Detection Limit
Cesva	SC260	Lower Limit: 30 dBA
Sound Level Meter (SLM)		Upper Limit: 137 dBA
		Peak: 140 dB



Monitoring Locations Around AB-1 Site



Sampling Methodology & Sites

LOCATION -01	LOCATION-02	LOCATION-03	LOCATION-04	LOCATION-05
AB1 Near DOD	AB1 Near Library	CPCB Site - Set 1	CPCB Site - Set 2	Dept. of Science Bldg
06-Apr-2026 4:00 PM	06-Apr-2026 2:50 PM	27-Mar-2026 11:37 AM	27-Mar-2026 12:50 PM	06-Apr-2026 3:00 PM
Leq: 58.71 dBA	Leq: 55.96 dBA	Leq: 58.58 dBA	Leq: 64.84 dBA	Leq: 72.53 dBA

Measurement Protocol

Sampling Interval: 15-second readings	Period: Daytime — 11:37 AM to 4:00 PM
Duration per site: 60 minutes (1 hour)	Method: Instantaneous Sound Level Monitoring

All measurements correspond to daytime construction & academic hours. No nighttime data collected due to administrative constraints.

Site-by-Site Acoustic Findings

AB1 Near DOD

▲ CPCB Limit: 50 dBA

Leq: 58.71 dBA

Lmax: 73.8 | Lmin: 46.4 | L90: 48.10 | L10: 60.30

Fluctuating environment. Peaks above 60 dBA. Ambient level exceeds silence zone limit by 2–7 dBA even during quiet periods.

AB1 Near Library

▲ CPCB Limit: 50 dBA

Leq: 55.96 dBA

Lmax: 66.8 | Lmin: 49.3 | L90: 49.60 | L10: 59.15

Most critical silence-dependent facility on campus. Lmin barely below the 50 dBA limit — no effective quiet period for students.

CPCB Site – Set 1

▲ CPCB Limit: 50 dBA

Leq: 58.58 dBA

Lmax: 71.4 | Lmin: 50.7 | L90: 52.31 | L10: 60.58

Morning session. Serves as campus core reference point. Noise driven by traffic and midday construction activity.

CPCB Site – Set 2

▲ CPCB Limit: 50 dBA

Leq: 64.84 dBA

Lmax: 81.0 | Lmin: 49.6 | L90: 52.41 | L10: 63.88

Afternoon session. Lmax of 81 dBA violates both academic and industrial standards. Noise escalated +10 dBA vs morning.

Dept. of Science Bldg

▲ CPCB Limit: 50 dBA

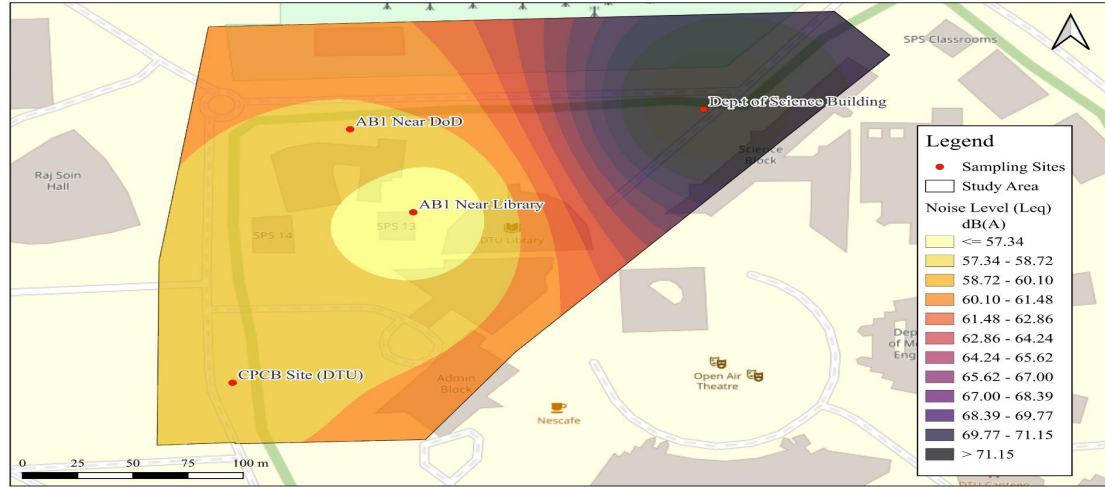
Leq: 72.53 dBA

Lmax: 90.2 | Lmin: 44.1 | L90: 45.02 | L10: 60.56

Most extreme readings in the study. 90.2 dBA outlier represents a real occupational hazard. Highly erratic acoustic profile ($\Delta = 46.1$ dBA).

Noise Contour Mapping

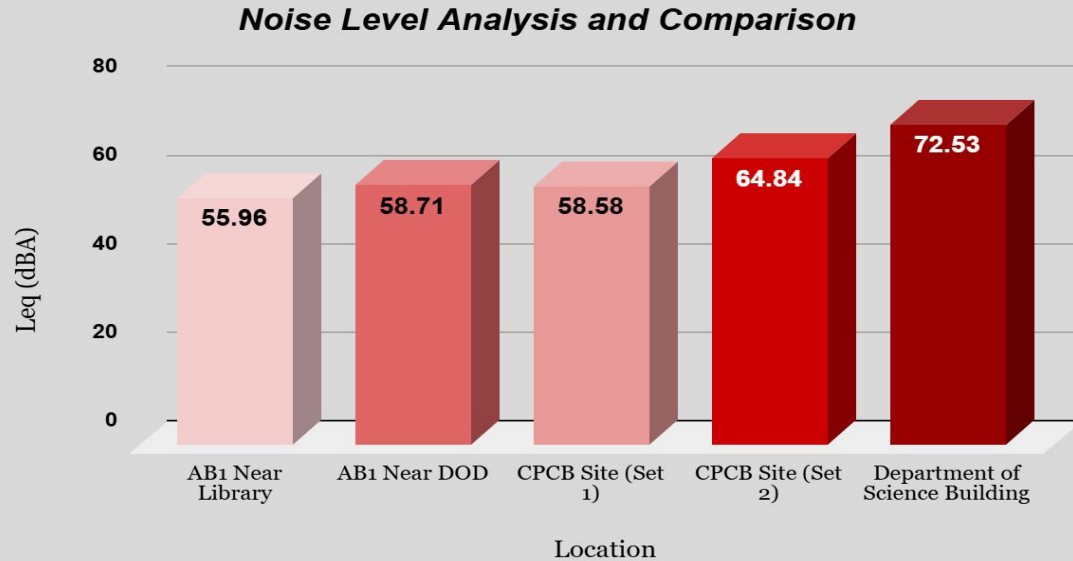
Noise Contour Map of the AB-1 Site in Delhi Technological University



Prepared by: EIA Survey Camp Group 4 (2025-2026),
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Delhi Technological University.
Date: 14/04/2026

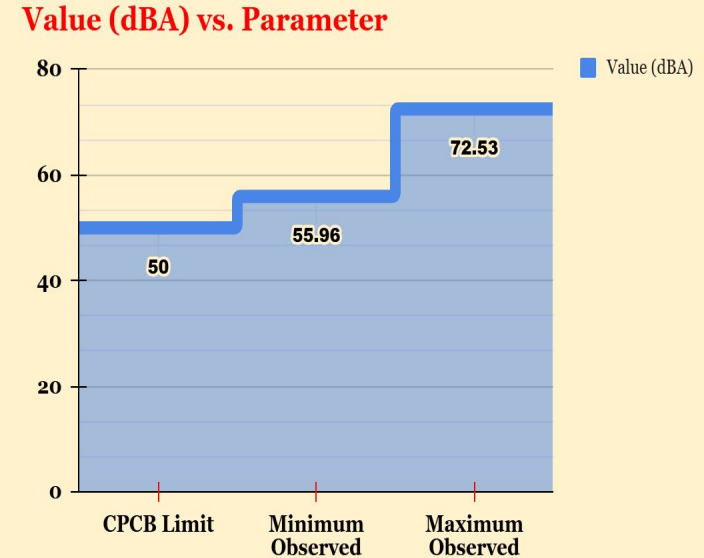
1. The noise contour map illustrates the spatial distribution of equivalent noise levels (Leq) across the AB-1 construction site at Delhi Technological University.
2. The map was generated using the **Inverse Distance Weighted (IDW) interpolation technique** in QGIS, based on noise measurements collected from multiple sampling locations.
3. The analysis indicates that noise levels range from approximately **57 dB(A)** to **above 71 dB(A)**. The **highest noise intensity is observed near the Department of Science Building**, while relatively lower levels are found around the **Library and Department of Design**.
4. This spatial variation highlights the uneven distribution of noise and identifies critical hotspots within the study area.

Noise Level Analysis and Comparison



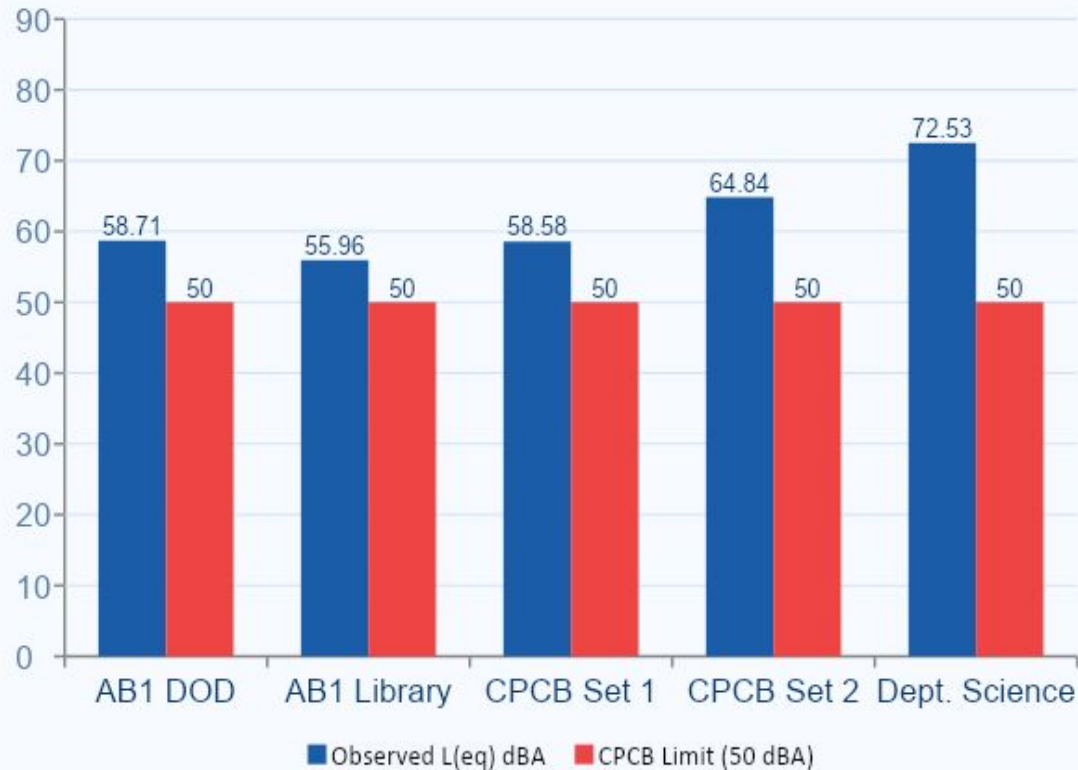
The analysis of equivalent noise levels (Leq) across all monitoring locations indicates that noise levels consistently exceed the CPCB permissible limit of 50 dB(A) for silence zones.

Among all locations, the **Department of Science Building** recorded the highest noise level (72.53 dB(A)), making it the most critical hotspot. The **CPCB Site (Set 2)** also shows significantly elevated levels, indicating increased noise during peak activity hours.



Relatively lower noise levels were observed near the **Library (55.96 dB(A))** and **Department of Design (58.71 dB(A))**, but these values still exceed the prescribed limits, highlighting the absence of a truly quiet zone within the campus.

Compliance Status Summary



Location	L(eq)	Dev.	Status
AB1 DOD	58.71	+8.71	FAIL
AB1 Library	55.96	+5.96	FAIL
CPCB Set 1	58.58	+8.58	FAIL
CPCB Set 2	64.84	+14.84	FAIL
Dept. Sci.	72.53	+22.53	FAIL

All 5 sites non-compliant

Max deviation: +40.2 dBA (Lmax)

CPCB Daytime Silence Zone Limit = 50 dBA

Primary Noise Sources & Impact Analysis

Construction Noise Sources

Traffic Noise

Movement of cars, bikes, DTU Golf Cart within campus.

Human Activities

Conversations, walking, social interactions.

Natural Sources

Wind, birds, and other environmental sounds.

Nearby Urban Influence

Noise from surrounding roads, Food Outlet (HIMS) **75–81 dBA**

Net increment: +10 to +25 dB → site levels reach 85–95 dBA

Impacts on Sensitive Receptors

Cognitive Performance

Noise above 55 dBA adversely affects reading comprehension, information processing, and exam scores.

Psychological & Physical Health

Exposure >70 dBA increases stress hormones; leads to hypertension, fatigue, and irritability.

Research & Lab Work

80–90 dBA vibrations interfere with precision instruments and spectroscopic analysis in Science labs.

Noise Mitigation Recommendations

Engineering Solutions

Acoustic Noise Barriers

3–4m high mass-loaded vinyl or acoustic panels around the full site perimeter.
Reduces noise by 10–15 dBA by breaking line of sight.

Acoustic Enclosures

All generators, compressors & pumps enclosed in soundproof housings.
Exhaust mufflers with ≥ 25 dBA insertion loss retrofitted on older machinery.

Low-Noise Technology

Electric concrete vibrators & modern low-noise equipment mandated for the contractor — reduces noise at the source.

Receptor Treatment

Secondary glazing and window sealing for the Central Library and Science labs to improve building sound attenuation.

Administrative Controls

Strategic Scheduling

High-noise tasks (drilling, demolition) prohibited during peak academic hours and examination periods. Schedule for early morning or late afternoon.

Equipment Positioning

Loudest machines placed as far from the Library and hostels as possible.
Exhaust outlets oriented away from academic buildings.

Mandatory Worker PPE

All workers in zones > 85 dBA must use hearing protection (earplugs or earmuffs) to prevent occupational hearing loss.

Real-Time Monitoring

Permanent noise monitors at the Library & Science Building, linked to a public web portal for transparency and regulatory accountability.

Conclusions & Key Findings

1

Universal Non-Compliance:

All 5 monitored locations violate the CPCB 50 dBA Silence Zone limit by margins of 6 to 23 dBA (L_{eq} basis).

2

Critical Hotspot:

Dept. of Science Building recorded the highest L_{max} of 90.2 dBA — a significant environmental and occupational hazard.

3

No Quiet Recovery:

At the Library and DOD, minimum noise levels are near or above the permissible average, leaving no restorative quiet period.

4

Temporal Escalation:

CPCB site data confirms noise escalates from 58.58 to 64.84 dBA between late morning and afternoon — linked to traffic and increased activity.

5

Legal Exposure:

DTU and contractors face significant regulatory liability. Violations >10 dBA require immediate intervention under Noise Pollution Rules, 2000.

